

Exercise - 1:

- a. Write a JAVA program to display default value of all primitive data type of JAVA
- b. Write a java program that display the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on value of D, describe the nature of root.

1. Displaying default value of all primitive data types

AIM: To write a JAVA program to display default value of all primitive data type of JAVA

SOURCE-CODE:

```
class defaultdemo
{
    static byte b;
    static short s; static int i;
    static long l;
    static float f;
    static double d;
    static char c;
    static boolean bl;
    public static void main(String[] args)
    {
        System.out.println("The default values of primitive data types are:");
        System.out.println("Byte :"+b);
        System.out.println("Short :"+s);
        System.out.println("Int :"+i);
        System.out.println("Long :"+l);
        System.out.println("Float :"+f);
        System.out.println("Double :"+d);
        System.out.println("Char :"+c);
        System.out.println("Boolean :"+bl);
    }
}
```

OUT-PUT:

The default values of primitive data types are: Byte :0

Short :0

Int :0

Long :0

Float :0.0

Double :0.0 Char : Boolean :false

b Roots of a quadratic equation

AIM: To write a java program that display the roots of a quadratic equation $ax^2+bx=0$.

Calculate the discriminate D and basing on value of D, describe the nature of root.

SOURCE-CODE:

```
import java.util.*; class quadraticdemo
{
    public static void main(String[] args)
    {
        int a, b, c; double r1, r2, D;
        Scanner s = new Scanner(System.in);
        System.out.println("Given quadratic equation:ax^2 + bx + c");
        System.out.print("Enter a:");
        a = s.nextInt();
        System.out.print("Enter b:");
        b = s.nextInt();
        System.out.print("Enter c:");
        c = s.nextInt();
        D = b * b - 4 * a * c; if(D > 0)
        {
            System.out.println("Roots are real and unequal");
            r1 = ( - b + Math.sqrt(D))/(2*a);
            r2 = (-b - Math.sqrt(D))/(2*a);
            System.out.println("First root is:"+r1);
            System.out.println("Second root is:"+r2);
        }
        else if(D == 0)
        {
            System.out.println("Roots are real and equal");
```

```

r1 = (-b+Math.sqrt(D))/(2*a);
System.out.println("Root:"+r1);
}
else
{
System.out.println("Roots are imaginary");
}
}
}

```

OUT-PUT:

Given quadratic equation: $ax^2 + bx + c$ Enter a:2

Enter b:3 Enter c:1

Roots are real and unequal First root is:-0.5

Second root is:-1.0

Exercise - 2

- a. Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
 - b. Write a JAVA program to sort for an element in a given list of elements using bubble sort
 - c. Write a JAVA program using String Buffer to delete, remove Character.
- Implementation of Binary search mechanism

2 a. Write a JAVA program to search for an element in a given list of elements using binary search mechanism.

AIM: To write a JAVA program to search for an element in a given list of elements using binary search mechanism

SOURCE-CODE:

```
import java.util.Scanner;

class binarysearchdemo
{
    public static void main(String args[])
    {
        int n, i, num, first, last, middle;
        int a[ ]=new int[20];
        Scanner s = new Scanner(System.in);
        System.out.println("Enter total number of elements:");
        n = s.nextInt();
        System.out.println("Enter elements in sorted order:");
        for (i = 0; i < n; i++)
            a[i] = s.nextInt();
        System.out.println("Enter the search value:");
        num = s.nextInt();
        first = 0; last
        = n - 1;
        middle = (first + last)/2; while( first <= last )
        {
            if ( a[middle] < num )first
            = middle + 1;
            else if ( a[middle] == num )
            {
                System.out.println("number found"); break;
            }
            else
            {
```

```

last = middle - 1;
}
middle = (first + last)/2;
}
if ( first > last )
System.out.println( " Number is not found");
}
}

```

OUT-PUT:

Enter total number of elements:5 Enter elements:

2 4 6 8 9

Enter the search value:8 number found

2b. Bubble sort

AIM: To write a JAVA program to sort for an element in a given list of elements using bubble sort

SOURCE-CODE:

```

import java.util.Scanner;

class bubbledemo
{
    public static void main(String args[])
    {
        int n, i,j, temp;
        int a[ ]=new int[20];

        Scanner s = new Scanner(System.in);

        System.out.println("Enter total number of elements:");
        n = s.nextInt();
    }
}

```

```

System.out.println("Enter elements:");
for (i = 0; i < n; i++)
a[i] = s.nextInt();
for(i=0;i<n;i++)
{
for(j=0;j<n-1;j++)
{
if(a[j]>a[j+1])
{
temp=a[j]; a[j]=a[j+1]; a[j+1]=temp;
}
}
}
System.out.println("The sorted elements are:");
for(i=0;i<n;i++)
System.out.print("\t"+a[i]);
}
}

```

OUT-PUT:

Enter total number of elements:

10

Enter elements:

3 2 5 7 6 8 9 1 4 0

The sorted elements are:

0 1 2 3 4 5 6 7 8 9

2. Merge sort:

AIM: To write a JAVA program to sort for an element in a given list of elements using merge sort

SOURCE-CODE:

```
import java.util.*;

class mergedemo
{
    public static void main(String args[])
    {
        int n1,n2,i,j,k;
        int a[ ]=new int[20];
        int b[ ]=new int[20];
        int c[ ]=new int[20];
        Scanner s = new Scanner(System.in);
        System.out.println("Enter number of elements in first array:");
        n1 = s.nextInt();
        System.out.println("Enter sorted elements of first array:");
        for (i
        = 0; i < n1; i++)
            a[i] = s.nextInt();
        System.out.println("Enter number of elements in second array:");
        n2 = s.nextInt();
        System.out.println("Enter sorted elements of second array:");
        for (j = 0; j < n2; j++)
            b[j] = s.nextInt();
        i = 0;
        j = 0;
        k = 0;
        while((i < n1) && (j < n2))
```

```

{
if(a[i] > b[j])
c[k++] = b[j++];
else
c[k++] = a[i++];
}
while(i < n1) c[k++] = a[i++];
while(j < n2) c[k++] = b[j++];
System.out.println("After merging the elements are:\n");
for(i = 0; i < (n1 + n2); i++)
System.out.print("\t"+c[i]);
}
}

```

OUT-PUT:

Enter number of elements in first array:

6

Enter elements of first array:

8 9 12 13 15 18

Enter number of elements in second array:

5

Enter elements of second array:

6 7 10 11 20

After merging the elements are:

6 7 8 9 10 11 12 13 15 18 20

2c. Implementing StringBuffer

AIM: To write a JAVA program using StringBuffer to delete, remove character

SOURCE-CODE:

```
class stringbufferdemo
{
    public static void main(String[] args)
    {
        StringBuffer sb1 = new StringBuffer("Hello World");
        sb1.delete(0,6);
        System.out.println(sb1);
        StringBuffer sb2 = new StringBuffer("Some Content");
        System.out.println(sb2);
        sb2.delete(0, sb2.length());
        System.out.println(sb2);
        StringBuffer sb3 = new StringBuffer("Hello World");
        sb3.deleteCharAt(0);
        System.out.println(sb3);
    }
}
```

OUT-PUT:

World

Some Content Hello World